Step 7 : Decision

i) Calculated $F_{0t} = 5.09 < f_{(2,\ 6),0.05} = 5.1456$, the null hypothesis is not rejected and we conclude that there is significant difference in the mean sales among the three programs.

ii) Calculate $F_{0b} = 3.59 > f_{(3,\ 6),0.05} = 4.7571$, the null hypothesis is rejected and conclude that there does not exist significant difference in the mean sales among the four salesmen.

Example 3.7

The illness caused by a virus in a city concerning some restaurant inspectors is not consistent with their evaluations of cleanliness of restaurants. In order to investigate this possibility, the director has five restaurant inspectors to grade the cleanliness of three restaurants. The results are shown below.

Inspectors	Restaurants		
	I	II	III
1	71	55	84
2	65	57	86
3	70	65	77
4	72	69	70
5	76	64	85

Carry out two-way ANOVA at 5% level of significance.

Solution:

Step 1 :

Null hypotheses

$H_{0I} : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$ (For inspectors - Treatments)

That is, there is no significant difference among the five inspectors over their mean cleanliness scores

$H_{0R} : \mu_{\text{I}} = \mu_{\text{II}} = \mu_{\text{III}}$ (For restaurants - Blocks)

That is, there is no significant difference among the three restaurants over their mean cleanliness scores

Alternative Hypotheses

H_{1I}: At least one mean is different from the other among the Inspectors

H_{1R}: At least one mean is different from the other among the Restaurants.

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